

Write a Prolog Program for STUDENT-TEACHER-SUB-CODE.

Aim:

To find the student teacher code

ALGORITHM

- Start
- Define the facts for students, teachers, and subjects with their corresponding codes.
- Define a rule to relate a student with a teacher and subject.
- Store the student name and student code.
- Store the teacher name and teacher code.
- Store the subject name and subject code.
- Define a rule student_teacher_subject/3 to find the teacher and subject associated with a student.
- Accept the required student as a query.
- Prolog searches the facts and applies the rule using unification and backtracking.
- Display the corresponding student, teacher, and subject codes.
- Stop.

Program:

% define some facts about teachers and their subject codes

teaches(john, math).

teaches(jane, english).

teaches(bob, science).

teaches(sue, history).

teaches(tom, art).

% define some facts about students and the subjects they are taking
takes(alice, math).

takes(alice, science).

takes(bob, english).

takes(bob, science).

takes(carol, history).

takes(carol, art).

takes(dave, math).

takes(dave, english).

takes(dave, art).

% define a predicate to look up the subject codes taught by a teacher

teaching_subjects(Teacher, Subject) :-

teaches(Teacher, Subject).

% define a predicate to look up the students taking a given subject

taking_students(Subject, Student) :-

takes(Student, Subject).

% example query: find the subjects taught by john

?- teaching_subjects(john, Subject).

Subject = math.

% example query: find the students taking science

?- taking_students(science, Student).

Student = alice ;

Student = bob.

% example query: find the students taking math and art

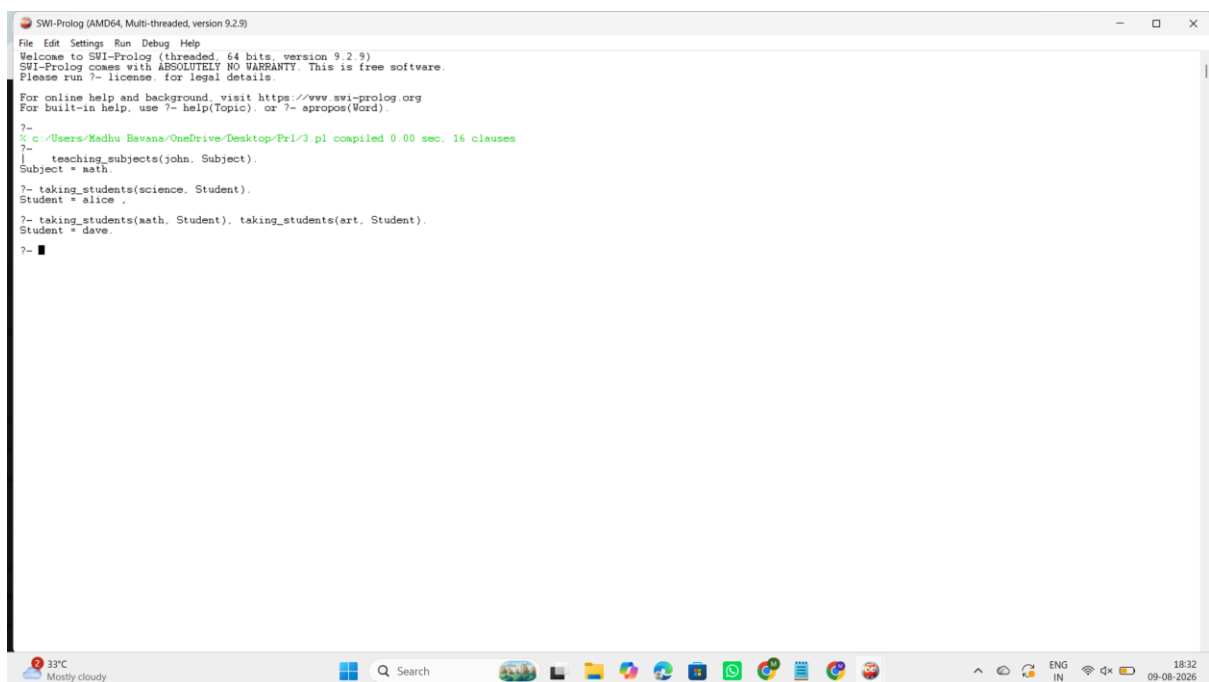
?- taking_students(math, Student), taking_students(art, Student).

Output:

Student = alice ;

Student = dave ;

Student = carol.



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SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/Madhu Bavana/OneDrive/Desktop/Pr1/3.pl compiled 0.00 sec, 16 clauses
?- teaching_subjects(john, Subject).
Subject = math.
?- taking_students(science, Student).
Student = alice.
?- taking_students(math, Student), taking_students(art, Student).
Student = dave.
?-
```

Result:

Hence Prolog Program for STUDENT-TEACHER-SUB-CODE is done.